

Karthikeyan S A

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PROFESSIONAL SUMMARY

A developer with strong fundamentals in object-oriented design, data structures, algorithms, and complexity analysis, backed by a background in Artificial Intelligence and Data Science. Skilled in Python-based development, REST API design, distributed system concepts, and relational database management using PostgreSQL. Gained hands-on backend development experience through an internship, focused on writing clean, maintainable code for production systems.

TECHNICAL SKILLS

Languages: Python, SQL, Linux, NOSQL, HTML5, CSS3, JavaScript
Frameworks & Libraries: Django, Django REST Framework, Bootstrap, React.js
Databases: PostgreSQL, MongoDB
Tools & Platforms: Git, GitHub, Docker, Cursor, GitHub Copilot

WORK EXPERIENCE

Backend Developer Intern

Sep 2024 – Nov 2024

Learn Basics

Namakkal, Tamil Nadu

- Collaborated with senior engineers to design and build Python/Django backend features for a production web application in an agile development environment.
- Implemented fault-tolerant error handling, structured logging, and performance optimizations to improve system stability and application efficiency.
- Applied object-oriented design principles and algorithm optimization techniques to resolve performance bottlenecks and fix critical bugs.
- Maintained structured Git-based version control workflows ensuring code quality and team collaboration.

PROJECTS

PlacementTrackDesk | Django, Python, PostgreSQL, Redis, Cloudinary

Mar 2026

- Reduced query load by implementing Redis caching on frequently accessed dashboard views.
- Replaced manual placement tracking by building a task assignment platform where admins post coding tasks and students submit proof of completion.
- Secured multi-user access by implementing role-based control (Admin/Student) with image proof uploads stored via Cloudinary.
- Improved admin visibility by building an analytics dashboard tracking real-time task completion rates across the student batch.
- Ensured data integrity by designing a normalized PostgreSQL schema with deadline enforcement and status tracking (Pending / In Progress / Completed).

Exam Dupe Shield | Python, OpenCV, ImageHash, Discord API, Minio API

Oct – Nov 2024

- Reduced manual exam review effort by automating image-based answer sheet comparison across all student submissions.
- Detected duplicate submissions by implementing perceptual hashing (ImageHash whash) to compare answer sheet images pairwise.
- Flagged similarities above 85% threshold as duplicate and below as unique, generating structured CSV reports for teacher review.
- Scaled file handling by integrating MinIO cloud storage for fault-tolerant upload and retrieval of answer sheet images.

EDUCATION

K S Rangasamy College of Technology

Aug 2022 – Apr 2026

B.Tech – Artificial Intelligence and Data Science | CGPA: 7.78

Namakkal, Tamil Nadu

Cheran Matric Hr Sec School

2020 – 2022

HSC – 83.6% | SSLC – 80.3%

Karur, Tamil Nadu

CERTIFICATIONS

Basics of Python – Infosys Springboard

Programming Fundamentals using Python – Part 1 & Part 2 – Infosys Springboard